container run via Testcontainers.

Before starting, we need to add new dependencies on Spring Security Test and Testcontainers Keycloak. Open the build.gradle file for the Catalog Service project (catalog-service), and update it as follows. Remember to refresh or reimport the Gradle dependencies after the new addition.

Listing 12.11 Adding dependencies to test Spring Security and Keycloak

```
ext {
  ...
  set('testKeycloakVersion', "2.3.0")                    ❶
}

dependencies {
  ...
  testImplementation 'org.springframework.security:spring-security-test'
  testImplementation 'org.testcontainers:junit-jupiter'
  testImplementation "com.github.dasniko:
  ➥ testcontainers-keycloak:${testKeycloakVersion}"     ❷
}
```

❶ The version of Testcontainers Keycloak

❷ Provides Keycloak testing utilities on top of Testcontainers

TESTING SECURED REST CONTROLLERS WITH @WEBMVCTEST AND SPRING SECURITY

First, let's update the `BookControllerMvcTests` class to cover the new scenarios, depending on the user authentication and authorization. For example, we could write test cases for `DELETE` operations in these situations:

- The user is authenticated and has the `employee` role.